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ABSTRACT

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A modified g_m/I_D design methodology for deeply scaled CMOS technologies

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Abstract This paper presents an updated version of the g_m/I_D -based sizing methodology for advanced short-channel CMOS technologies. The objective of this technique is to quickly and accurately size any linear analog circuit, top-down, from some required specifications and evaluate the remaining ones. A database describing the underlying MOS technology is taken as input of the sizing script, making the sizing process technology and corner independent. An advanced CMOS technology is analyzed, underlining the limitations of the original g_m/I_D methodology and its past improvements, then the proposed methodology is described in detail and tested successfully on a double stage amplifier, using two different CMOS technologies in all process-voltage-temperature corners.

Keywords CMOS · Analog design · Design reuse · CAD methodology · Transistor sizing · Ultra-deep-submicron · g_m/I_D

1 Introduction

Contrary to digital gate design, where the design space is relatively narrow, what makes the difficulty of analog cell design is its large amount of degrees of freedom. The goal of an analog sizing technique is to come up with the sizes of the transistors composing an integrated circuit from decisive technology data, given specifications and for a

certain architecture (Fig. 1). This top-down aspect is usually completed with the ability to evaluate the circuit parameters from the transistor sizes, like a circuit simulator.

Several methodologies, more or less automated, were developed so far to reach this goal. They evolved with the underlying technology, so as to take into account the physical effects of more complex fabrication processes. These methodologies however do not aim at replacing the designer, but assisting him in his task [1–4, 8].

The most crude approach consists in making SPICE simulations of a circuit using sweeps on independent widths (W) and lengths (L). From the sensitivity of the circuit characteristics to the size changes, with time and design experience, the designer can find sizes respecting the specifications.

Many other approaches were developed using first-order equations or SPICE-generated look-up tables to evaluate a starting point for the widths of the transistors [1], or refine the results using numerical optimization techniques [2–4]. Although they were rarely maintained, many design softwares were developed from these methodologies [5, 6]. Gielen [7] in 2000 made a review of the sizing tools and approaches at the time, regardless of the methodology these tools used.

The so-called g_m/I_D design methodology was introduced in [8] initially in order to help sizing micrometer-long transistors in analog circuits so that designers could benefit of all MOSFET levels of inversion.

A SPICE simulator determines the electrical state of a single MOS transistor from the access voltages and transistor dimensions (V_{GS} , V_{DS} , W and L), which can only be linked to the specifications via complex equations, models or simulations. On the contrary, the original g_m/I_D methodology only takes as input three independent parameters:

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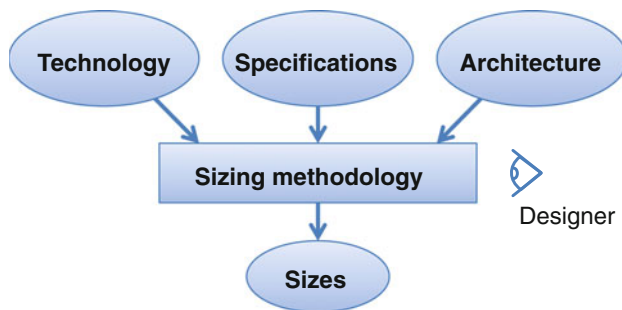


Fig. 1 Principle of an analog sizing technique

g_m/I_D , L and I_D , which are more significant for a designer since their values are directly inferred by specifications. However, they may not be sufficient to accurately describe the state of a very short channel transistor as we will show.

For advanced technologies, the authors consider that four parameters (g_m/I_D , V_{DS} , L and I_D) have to be taken into account to have a sufficient accuracy on the circuit performance estimation. The proposed methodology aims at determining them from circuit level characteristics and the circuit architecture. Then with a data table of the normalized current function of vectors of g_m/I_D , V_{DS} and L , for a given bias current, having the value of these three parameters is equivalent to knowing the transistor sizes or reciprocally.

In this paper, we will sum up the g_m/I_D methodology evolution so far (Sect. 2) Then by analyzing the characteristics of a CMOS technology with minimum feature size of 60 nm, we will see how the evolution of CMOS technology imposes an improvement of this design methodology (Sect. 3) From the set of relations extracted of this simulation-based analysis, we shall build up an updated sizing methodology (Sect. 4), that we will test on a double-stage amplifier and evaluate its effectiveness comparing its results with SPICE simulations (Sect. 5) And finally, Sect. 6 will show how effective the methodology is considering corners and technology transfer.

2 Standard g_m/I_D methodology

The main concept is to link transistors features and circuit-level performances via one key characteristic: the gate transconductance efficiency (g_m/I_D) versus the drain current normalized by the transistor aspect ratio ($I_N = I_D/(W_{EFF}/L_{EFF})$, with W_{EFF} and L_{EFF} , respectively the transistor effective width and length) [8, 9]. This characteristic establishes a bijection between the gate transconductance (g_m) and the transistor aspect ratio for a given drain current. It was supposed constant in saturation, whatever the drain voltage or size of the considered MOSFET.

The standard small-signal equivalent circuit of a single transistor (Fig. 2) is completed by its output conductance

(g_d) linked to the early voltage, assumed proportional to the gate length and independent of the bias ($V_{EA} = (g_d/I_D)^{-1} \approx V'_{EA} \cdot L$, with a constant V'_{EA} in V/ μm).

These characteristics are either extracted from a simulation, or from an actual MOS characterization. Therefore the designer can easily compute the circuit small-signal parameters and link them to the transistor sizes. For example, the DC gain (A_{V_0}) and transition frequency (f_t) of a common-source transistor loaded by a capacitor C_L are:

$$A_{V_0} = \frac{g_m}{g_d} = \frac{g_m}{I_D} \cdot V'_{EA} \cdot L \quad (1)$$

$$f_t = \frac{g_m}{2\pi \cdot C_L} = \frac{g_m}{I_D} \cdot \frac{I_D}{2\pi \cdot C_L} \quad (2)$$

In a top-down approach, i.e. from specifications to design solution, from a given A_{V_0} or the $\{f_t, C_L\}$ couple and a fixed L , g_m/I_D can be found and then, through the g_m/I_D versus I_N characteristic (Fig. 3), W .

Earlier sizing methodologies used independent W s as design variables. Yet the mapping of a particular MOSFET W to circuit performances has high interactions with W s and L s of other MOSFETs and many useless regions of design space (see Fig. 3). On the contrary, the biggest benefit of using g_m/I_D as main design variable is that it has weak interactions in the mapping. It facilitates rapid sizing and exploration of the design space.

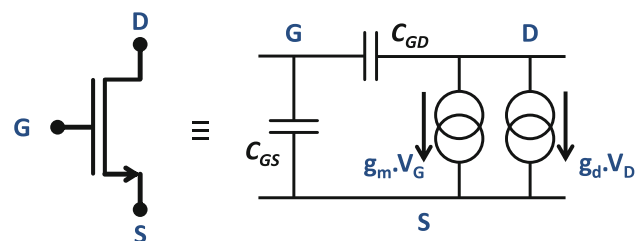


Fig. 2 Standard small-signal equivalent schematic of a MOS transistor with V_G , V_S , V_D referred to the bulk

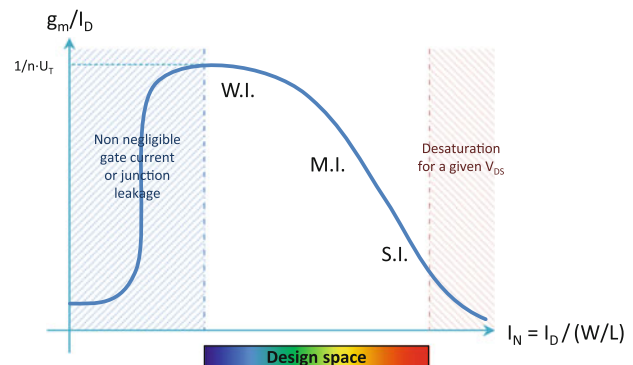


Fig. 3 Design space using the g_m/I_D sizing methodology, reduced to an area meaningful from a circuit point of view. W.I., M.I. and S.I. respectively are weak, moderate and strong inversion

Aside g_m/I_D and V_{EA} , the designer can also use simple equations to approximate large-signal and DC characteristics and finally come up with biases and transistor dimensions ensuring the required specifications.

These relations however are based on long-channel MOSFET models, and this methodology was developed for technologies and designs with supply voltages of 2.5 V or more. Its objective was to get a quick, technology independent top–down sizing to be possibly refined by simulation afterwards. Some authors [10–12] attempted to improve this methodology but their studies are based on the EKV model [13] adapted to a particular technology, while EKV model cards are not supplied by all foundries. Furthermore, these works did not address neither rapid portability, nor sub-100 nm technologies. Murmann [14] updated the g_m/I_D methodology successfully for more advanced technology down to 180 nm but considers in his papers that V_{DS} effects are negligible. The attempts of porting a design automatically using his technique to another technology are also limited [15].

As will show the next section, for sub-100 nm technologies, the transistor characteristics a designer needs to build up a circuit respecting its specifications are strongly affected by V_{DS} and L .

3 Significant parameters affecting short-channel transistors

We shall aim in this section at determining a reduced set of parameters from which we can accurately describe a single transistor in a circuit, and infer all other parameters. This reduced set will serve as input to a data table containing all the MOS characteristics required by the designer of a linear analog circuit.

This section introduces the most important characteristic curves from a circuit designer point of view of a standard- V_T NMOSFET in saturation from a 60 nm, 1.2 V bulk CMOS technology, low power flavor. The minimum size transistor in this particular technology is 120 nm wide by 60 nm long. The I–V curves are extracted from a BSIM4 SPICE model of a 1.2 μm wide by 60 nm or 6 μm long transistor, sweeping the gate-source voltage and keeping the drain-source voltage at 0.2, 0.5 or 1.0 V, unless stated otherwise. These characteristics will be extracted into Matlab to describe the technology in the methodology.

3.1 Gate transconductance efficiency— g_m/I_D

3.1.1 Dependence to drain-source voltage and channel length

Figure 4 presents the well-known g_m/I_D versus I_N characteristic. The solid curves, for 6 μm MOSFETs, tend to

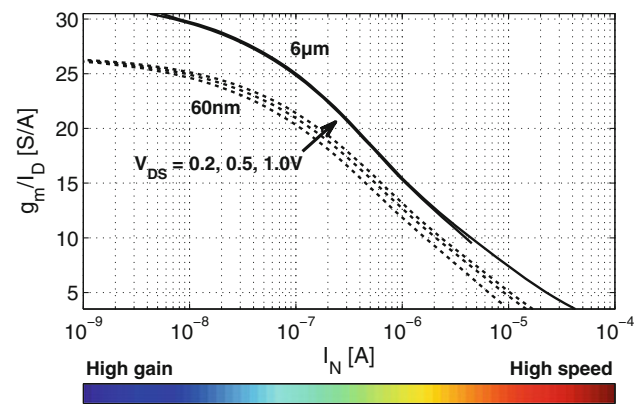


Fig. 4 g_m/I_D versus I_N for $L = 60$ nm (dotted) and 6 μm (solid) and $V_{DS} = 0.2, 0.5, 1.0$ V, in saturation

confirm a reduced sensitivity to the drain bias voltage of this characteristic for long transistors. For shorter transistors though, the curve shifts downwards and shows strong drain-source voltage related effects. The presumption on which the original g_m/I_D method is based on is thus no longer valid.

Nevertheless, as Foty underlines in [16], knowing g_m/I_D gives the designer an idea of the inversion level, and thus the location of the transistor on the high gain versus high bandwidth ladder. Having conductances normalized by the drain current also permits the designer to make a complete current-independent sizing. Compared with a sizing based on the width, g_m/I_D has more open, freer interactions with circuit performances, allowing easier exploration of the design space. The transconductance efficiency therefore remains a central trait for analog design.

3.1.2 Dependence to channel width

Although lesser than the drain-source voltage (V_{DS}) and the gate length, the transistor width also has an influence upon the g_m/I_D versus I_N characteristic.

The characteristic presented in Fig. 4 is extracted for a reference width W_{REF} of 1.2 μm . The BSIM model-card of the technology we are considering ranges between $W_{MIN} = 120$ nm and $W_{MAX} = 12$ μm . We can extract the same characteristic for all W in this range and for each of them, estimate the error made on I_N compared to $I_{N,REF}$, the value extracted with W_{REF} for the same g_m/I_D and L . The drain-source voltage having only a limited influence on this error, it is not taken into account.

The error on I_N is plotted in Fig. 5. As it can be relatively large, we may want to optimize the reference width so that smaller ranges of widths significantly impact the characteristic.

Sweeping W_{REF} from W_{MIN} to W_{MAX} we come up with Fig. 6. This figure shows the widths range of the design

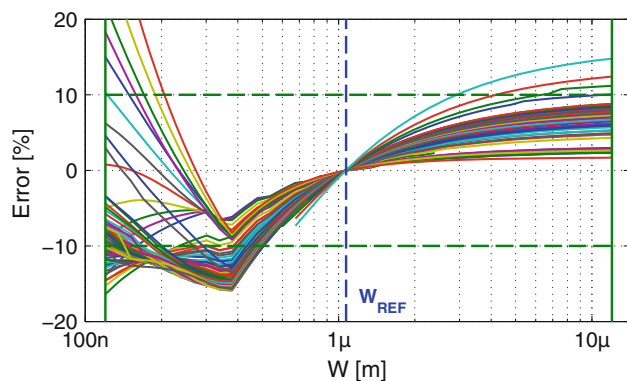


Fig. 5 Relative error between the reference normalized current, extracted for a given $W_{REF} = 1.2 \mu\text{m}$, and the normalized current for another W , versus W , for several L between 60 nm and $6 \mu\text{m}$ and several g_m/I_D between 3 and 30 S/A, of a standard V_T

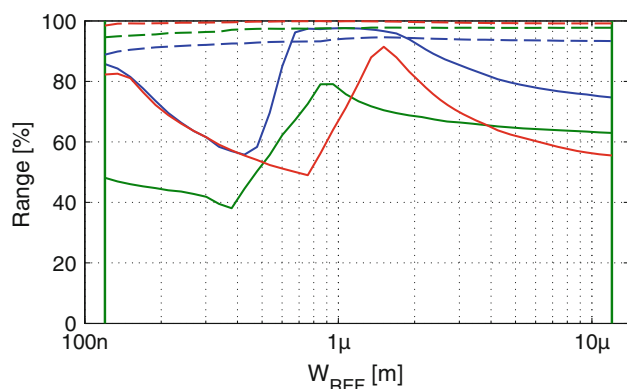


Fig. 6 Percentage of widths in the design space leading to a normalized current error of less than $\pm 10 \%$ versus the reference width, for NMOS (solid) and PMOS (dashed), low (blue), standard (green) or high (red) threshold voltage (Color figure online)

space changing I_N of less than $\pm 10 \%$ versus the chosen W_{REF} value, for each transistor type. The PMOSFETs of this technology being almost insensible to width narrowing, a middle range W_{REF} is suitable for all threshold voltages. For NMOS transistors, the optimum values are $W_{REF, lowVt} = 1.08 \mu\text{m}$, $W_{REF, stdVt} = 960 \text{ nm}$ and $W_{REF, highVt} = 1.5 \mu\text{m}$ to maximize the usable width design range. We shall use these values thereafter.

The g_m/I_D and L leading to large errors are extreme values, that are less likely to be used in a circuit.

3.2 Drain conductance to current ratio— g_d/I_D

In the early days of the g_m/I_D methodology, the Early voltage ($V_{EA} = (g_d/I_D)^{-1}$) was considered, as a first approximation, proportional to the gate length, but Binkley [12] pointed out that, because of HALOs and pocket implants of sub-micrometer technologies, it could be

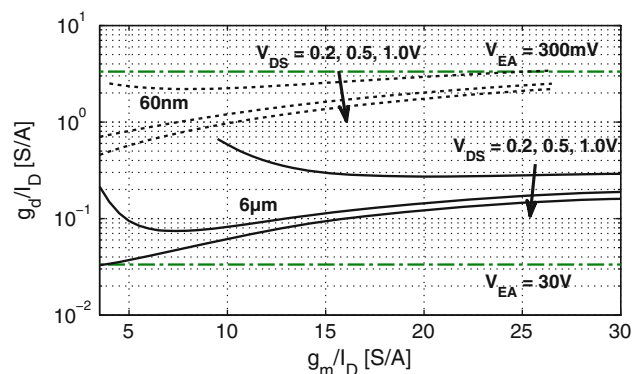


Fig. 7 g_d/I_D versus g_m/I_D for $L = 60 \text{ nm}$ (dotted) and $6 \mu\text{m}$ (solid) and $V_{DS} = 0.2, 0.5, 1.0 \text{ V}$, in saturation. In axis line, the same for a constant $V_{EA} = 5 \text{ V}/\mu\text{m}$

considered as such only for a limited range of lengths, and lengthening the gate to significantly improve the gain is no longer an option beyond this range.

Figure 7 presents in axis line the g_d/I_D for short (60 nm, $V_{EA} = 300 \text{ mV}$) and long ($6 \mu\text{m}$, $V_{EA} = 30 \text{ V}$) transistors considering a constant V_{EA} of $5 \text{ V}/\mu\text{m}$, and in solid and dotted lines the actual simulated value versus g_m/I_D , V_{DS} and L . It illustrates a strong non-linear dependence of g_d/I_D with those three parameters so that the constant V_{EA} approximation clearly underestimates g_d/I_D for long transistors and overestimates it for short ones. An accurate estimation of an analog gain or of a pole frequency thus necessitates knowing g_m/I_D , V_{DS} and L . V_{DS} contribution has not been taken into account before, though it can lead to errors as large as an order of magnitude.

3.3 Source transconductance efficiency— g_{ms}/I_D

If only the gate-source voltage were to influence the channel, the source transconductance ($g_{ms} = \partial I_D / \partial V_S$) would be exactly opposite to the gate transconductance. But a source voltage variation impacts the drain current via bulk-source and drain-source voltage variations as well. The substrate factor ($n = -g_{ms}/g_m = (g_m + g_{mb} + g_d)/g_m \approx g_{mb}/g_m + 1$, g_{mb} being the bulk transconductance) relating g_{ms} and g_m , at first supposed equal to one, then rather constant for micrometer technologies, actually varies much more for short transistors.

For micrometer transistors, g_d is negligible compared to g_m , the above approximation, neglecting g_d compared to g_m , remains valid and only the bulk effect degrades n (here $n = 1.16$). Short transistors exhibit a larger g_d that may become comparable to the other conductances and that will also influence the g_{ms} value. As was stated before, g_d is a function of g_m/I_D , V_{DS} and L , as is g_{mb} to a lesser extent. Still, the channel length and drain-source voltage have to be taken into account (see Fig. 8) if an accurate estimation

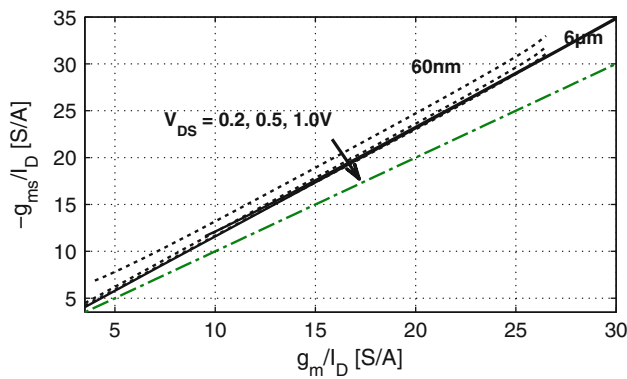


Fig. 8 $-g_{m}/I_D$ versus g_m/I_D for $L = 60$ nm (dotted) and $6\ \mu\text{m}$ (solid) and $V_{DS} = 0.2, 0.5, 1.0$ V, in saturation. In axis line, the same for $n = 1$

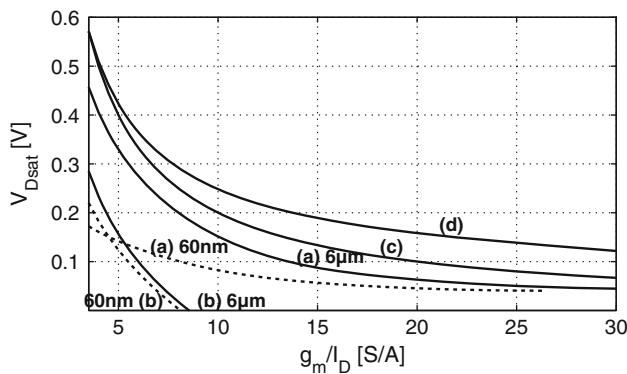


Fig. 9 V_{Dsat} versus g_m/I_D for $L = 60$ nm (dotted) and $6\ \mu\text{m}$ (solid) using four different methods: **a** BSIM, **b** GVO/ n , **c** $2/(g_m/I_D)$ and **d** EKV [13]

of the source transconductance is required. It is critical for voltage buffers and voltage shifters for example.

3.4 Drain to source saturation voltage— V_{Dsat}

Since the lowering of the maximum supply voltage in the latest technologies makes voltage swings a major constraint [17], accurately knowing V_{Dsat} actually allows the designer to set his own margins and to improve his optimization.

Figure 9 presents the several possibilities previously used to evaluate V_{Dsat} compared to the actual extraction of V_{Dsat} from the BSIM4 model (Fig. 9(a)). First from the gate voltage overdrive (GVO): $(V_{GS} - V_T)/n$ (Fig. 9(b)), n being the substrate factor as defined earlier and V_T extracted here using the $I_D/\sqrt{g_m}$ method [18], but this estimation does not prove convincing anymore. The two other methods, independent of L , i.e. the expression $2/(g_m/I_D)$ (Fig. 9(c)), and EKV for hand calculations (Fig. 9(d)) [13] fit rather well the long transistor V_{Dsat} behavior if adding a margin for design is considered, but fairly

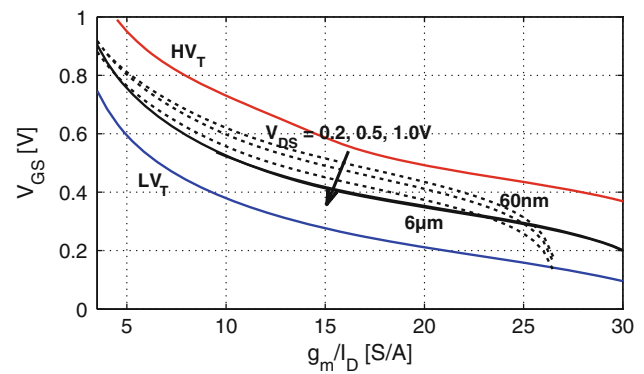


Fig. 10 V_{GS} versus g_m/I_D for $L = 60$ nm (dotted) and $6\ \mu\text{m}$ (solid) and $V_{DS} = 0.2, 0.5, 1.0$ V, for a standard- V_T NMOS in saturation; curves in blue and red, respectively for a low (LV_T) and high- V_T (HV_T) $6\ \mu\text{m}$ -long NMOS (Color figure online)

overevaluate it for short transistors. Only extracting the BSIM model values versus g_m/I_D and L leads to an accurate simulation to which we can add a margin if needed.

3.5 Gate to source voltage— V_{GS}

As can be seen on Fig. 10, the gate-source voltage relation with g_m/I_D is not invariant but depends on V_{DS} and L as well. Short transistors in strong and moderate inversion require a higher V_{GS} for the same inversion level than long transistors, while it is lower in weak inversion.

In a circuit and for a multi-threshold technology, a transistor threshold voltage will be chosen using this characteristic, from the DC point required on each node: for example, the blue (LV_T) and red (HV_T) lines correspond to low and high- V_T long transistors.

3.6 MOSFET parasitic capacitances— C_{ij}

The transistor parasitic capacitances are defined as $C_{ij} = k \cdot \frac{\partial Q_i}{\partial V_j}$, $k = +1$ if $i = j$, -1 otherwise. i and j are any of the drain (D), source (S), gate (G) or bulk (B) nodes. Each of these 16 non-reciprocal capacitances is the sum of two parallel sub-capacitors: an intrinsic capacitance due to the charges below the active gate, that can be normalized by the effective gate area (i.e. $W_{EFF} \times L_{EFF}$); and an extrinsic capacitance, due to the fringing fields through MOS geometry and diodes or between metal connections, that can be normalized by either W or L , depending on its location [19, 20].

The intrinsic capacitances depend on the inversion level, the drain-source voltage and the length. Three overlap and two junction capacitances form the five main extrinsic capacitances: C_{GB0} is normalized by L , C_{GD0} and C_{GS0} by W , C_{BDj} and C_{BSj} are function of V_{BD} and V_{BS} respectively and also normalized by W . All capacitances are

extracted versus the parameters they are function of, from BSIM simulations.

4 Sizing methodology

Given the major influence of the channel length and of the drain-source voltage on the characteristic parameters, we have to take these two parameters into account to improve the accuracy of a sizing methodology. Using the data extracted in the preceding section and knowing the optimal W_{REF} for each transistor type, the g_m/I_D methodology can be modified to precisely mimic the behavior of linear circuits designed in an advanced CMOS technology. We will develop the methodology for a single transistor: first see how to calculate W from g_m/I_D , V_{DS} and L and evaluate the error made. Then explain how to propagate the circuit performance parameters to set the transistors g_m/I_D , V_{DS} and L .

4.1 Data tables

The data tables describing the technology are extracted versus three vectors of g_m/I_D (between 1 and $1/(n \cdot U_t)$), V_{DS} (between 0 and 1 V) and L (between L_{MIN} and $100 \times L_{MIN}$) from the SPICE simulation of a W_{REF} -wide transistor, in the conditions of Sect. 3. A call to SPICE in the sizing script would have been far slower: loading the tables in Matlab takes only a few seconds the first time the script is run.

Choosing values for g_m/I_D , V_{DS} and L , we can look up from the data tables all the other characteristics of this particular transistor (i.e. g_d/I_D , g_{ms}/I_D , I_N , V_{GS} , V_{Dsat} , and the intrinsic capacitances, see Fig. 11) and deduce from them all sizes, capacitances and transconductances as a function of the bias current.

For example, using the data table used for Fig. 4 (g_m/I_D versus I_N), to a given $\{g_m/I_D|_A, V_{DS|A} \text{ and } L_A\}$ corresponds one and only one value of I_{N_A} , determined using a triple

interpolation at point A. Having I_{N_A} and I_{D_A} , the effective aspect ratio $W_{EFF}/L_{EFF}|_A$ is found, and then, W_A from L_A .

However, for the sizing to be correct, the transistor described in SPICE with $\{V_{DS}, V_{GS}, W \text{ and } L\}$ has to be the same as the one described with $\{g_m/I_D, V_{DS} \text{ and } L\}$ biased with I_D .

Table 1 sums up a comparison between the parameters estimation with Matlab and a SPICE simulation of a single MOS transistor. The Matlab estimation is based on the three parameters g_m/I_D , V_{DS} and L , the transistor being biased with a known current. The SPICE simulation for comparison is based on the size and the voltage bias, V_{DS} and V_{GS} . V_{GS} is adjusted after a first simulation in order to cancel the error on g_m/I_D , the errors listed in the table thus only lie in the imperfect link between $\{g_m/I_D, V_{DS}, L\}$ and all the assessed parameters, the interpolation being not as accurate as the SPICE model calculation.

For each case of Table 1, the estimation is computed from the first four values (g_m/I_D , V_{DS} , L and I_D). The results we come up with in Matlab are the next three values (W , V_{GS} and V_{Dsat}), plus the 4 conductances and 16 capacitances, the values of which are not listed for concision (ΔC_{mean} is the mean of all errors on the 16 capacitances estimation). The estimation errors between Matlab and SPICE for current independent, then current dependent parameters are listed afterward.

The error in the first column (with $W = W_{REF}$) proceeds only from the Matlab interpolations, and not the narrow

Table 1 Error on transistor characteristics between Matlab estimation and SPICE simulation

Case	1	2	3	4	5
g_m/I_D (S/A)	16	10	5	8	22
V_{DS} (V)	0.2	0.5	0.5	0.3	0.2
L (μm)	0.060	0.600	1.000	0.180	0.060
I_D (μA)	5	5	50	500	0.5
W (μm)	W_{REF}	0.797	2.637	18.77	0.631
V_{GS} (mV)	500	556	768	662	398
V_{Dsat} (mV)	54	135	301	143	43
ΔI_N (%)	0.1	1.1	6.6	8.8	2.4
ΔV_{GS} (%)	0.0	0.3	1.3	1.9	1.0
ΔV_{Dsat} (%)	0.9	2.7	1.2	0.9	1.2
$\Delta g_d/I_D$ (%)	1.3	0.1	0.4	0.9	2.0
$\Delta g_{ms}/I_D$ (%)	0.1	0.1	0.6	0.7	0.4
$\Delta g_{mb}/I_D$ (%)	0.1	0.8	4.5	6.3	2.5
ΔI_D (%)	0.1	1.1	6.6	8.8	2.4
Δg_m (%)	0.0	1.1	6.6	8.8	2.4
Δg_d (%)	1.4	1.2	6.1	7.8	4.3
Δg_{ms} (%)	0.2	1.2	7.2	9.6	2.8
Δg_{mb} (%)	0.0	1.9	11.4	15.6	4.8
ΔC_{mean} (%)	0.1	0.4	2.3	2.7	0.9

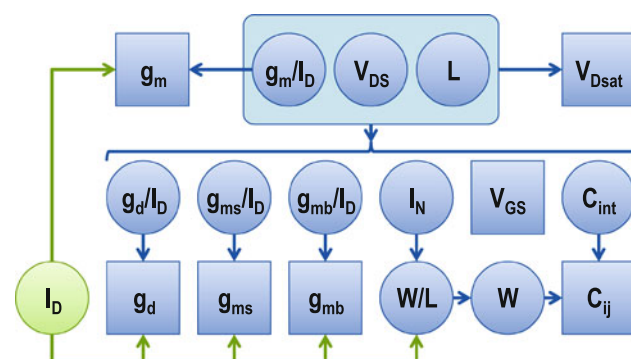


Fig. 11 Single transistor sizing diagram from g_m/I_D , V_{DS} and L

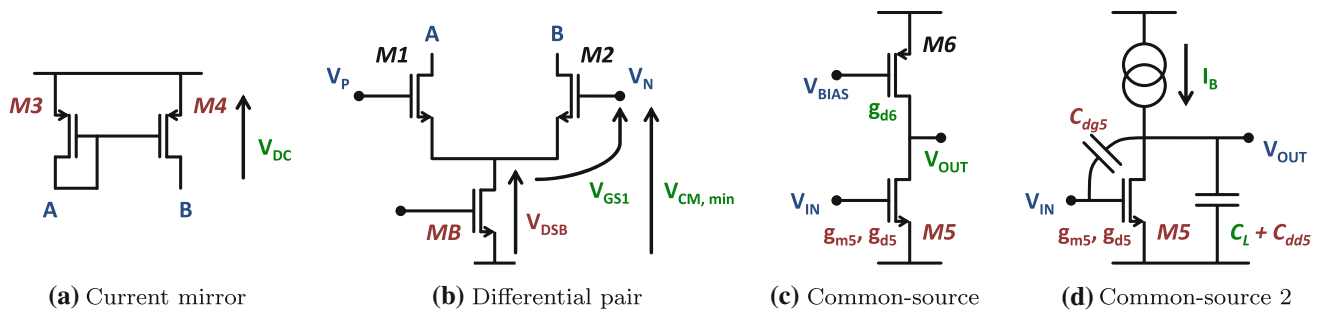


Fig. 12 Design entry points from DC **a**, large-signal **b**, **c** or small-signal **d** specifications; in green, specifications, in red, design results (Color figure online)

width effects. The other cases are representative of the main corners on g_m/I_D , V_{DS} and L .

The estimation of g_{mb} is deduced of the other conductances values $g_{mb} = -(g_m + g_d + g_{ms})$, the error on this transconductance may then be larger than the others. Still, this table shows a fair accuracy of the methodology so far, between SPICE and g_m/I_D representations.

4.2 Sizing principles

At circuit level, the idea is to determine, from design specifications, g_m/I_D , V_{DS} and L for each transistor, since all other parameters can be accurately derived from them. Generally, g_m/I_D is linked to a circuit performance, V_{DS} is set by the V_{GS} of another transistor, and L depends on the technology, though it can of course be optimized for a DC gain or a resistance to V_{DS} or corner variations.

Konishi [15], using Murmann's g_m/I_D approach [14], sets g_m/I_D and L using the transistor transition frequency and intrinsic gain only. This section introduces ways of sizing from DC and large signal specifications as well as small signal ones.

4.2.1 Sizing from a DC specification (V_{DC} , Fig. 12(a))

Considering the current mirror $M3$ – $M4$, for technologies with reduced Early voltage, a good current copy will only be obtained by equalizing both the sizes and the DC drain-source bias. We have seen in Fig. 10 that for a given V_{DS} and L , there is a bijection between the gate-source voltage and g_m/I_D . Knowing the DC voltage on node B, we can find a g_m/I_D so that $V_{DS_3} = V_{GS_3} = V_{DC}$. Thus from V_{DC} and setting the channel length according to other specifications, the transconductance efficiency of both transistors is obtained using an interpolation on the V_{GS} versus g_m/I_D characteristic.

For example, using low- V_T PMOS transistors of 600 nm channel length, a V_{DC} of 0.4 V yields a g_m/I_D of 12.67 S/A.

We may also choose more wisely the length of these transistors. Figure 13 plots the error on the copied current versus the output voltage for several channel lengths. Considering the

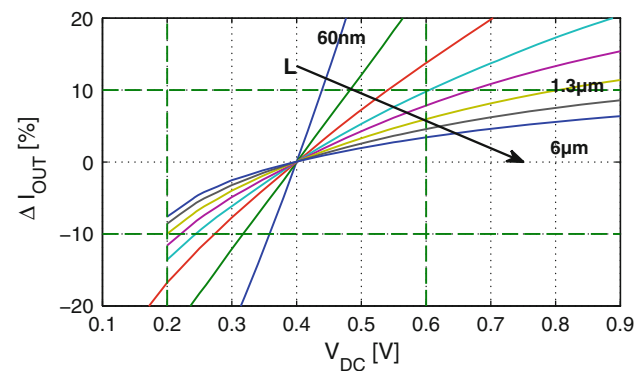


Fig. 13 Relative error on output current due to output voltage variation for several L values, from 60 nm to 6 μm , for a given $V_{GS_3} = 0.4\text{ V}$, versus V_{DC}

large signal output swing on this node, and deciding on a maximum current error, we can set the length of both transistors. Here for example, considering a swing of 0.2 V around the nominal voltage bias of 0.4 V, and a maximum current copy error of 10 %, the minimum required length is 1.3 μm . This result is independent of the drain current.

4.2.2 Sizing from a large-signal specification

Case 1 ($V_{CM, min}$, Fig. 12(b)) We are considering in this case the bias of the differential pair in Fig. 12(b), and as input for design, the minimum common-mode voltage $V_{CM, min}$. As long as the transistors $M1$ and $M2$ are correctly biased, that is unless the common-mode voltage goes below $V_{GS_1} + V_{DSat_B}$, desaturating MB , their g_m/I_D and then their gate-source voltage are unchanged (for $V_{SB} = 0$). We noticed earlier in Fig. 9 that for a given L , V_{DSat} is settled by setting g_m/I_D . So, if $M1$ and $M2$ were already sized in a preceding step from another specification, and V_{GS_1} is known, the g_m/I_D of MB is found from $V_{DSat_B} = V_{CM, min} - V_{GS_1}$. V_{DS} shall be deduced from the nominal common-mode voltage: $V_{DS_B} = V_{CM} - V_{GS_1}$.

For example, using a standard- V_T NMOS transistor of 180 nm channel length, having $V_{GS_1} = 0.25\text{ V}$ and an input DC voltage of 0.5 V, a minimum common-mode voltage of 0.32 V yields a g_m/I_D of 15.13 S/A.

Case 2 (V_{OUT} swing, Fig. 12(c)) The output voltage swing of the schematic Fig. 12(c) is commonly defined as the supply voltage minus $M5$ and $M6$ saturation voltages. But the major influence of V_{DS} , even in saturation, on the drain conductance makes the voltage gain of this common-source stage very dependent on the output voltage. We could therefore define the output swing as the voltage range where the DC gain remains above a required level.

Here,

$$A_{V_0} = \frac{g_{m5}}{g_{d5} + g_{d6}} = \frac{g_m}{I_D|_5} \cdot \left(\frac{g_d}{I_D|_5} + \frac{g_d}{I_D|_6} \right)^{-1} \quad (3)$$

which is, in the g_m/I_D domain, independent of the bias current. Considering $M6$ sized in a preceding step, and as $g_d/I_D|_5$ is a function of $g_m/I_D|_5$, we can compute a matrix of A_{V_0} as a function of g_m/I_D and V_{DS} , for a given L . From this matrix, we are able to deduce for each possible g_m/I_D a voltage range where $A_{V_0} > A_{V_{0,spec}}$. By comparing these range values to the desired swing, we can find a suitable g_m/I_D .

For example, considering a supply voltage of 1.0 V and for $M6$ a 600 nm-long standard- V_T PMOS with a g_m/I_D of 15 S/A, the whole stage DC gain can be computed as a function of $g_m/I_D|_5$ and the output DC voltage (see Fig. 14(a)). Targeting a DC gain of $A_{V_{0,spec}} = 28$ dB, an output voltage range ensuring this gain can be determined for each possible $g_m/I_D|_5$ value, as plotted Fig. 14(b). For a 0.5 V output swing, the minimum $g_m/I_D|_5$ is of 18.16 S/A.

The green axis line of the same figure shows the output voltage range where $M5$ and $M6$ stay in saturation: $V_{DD} - V_{Dsat5} - V_{Dsat6}$, as could be calculated as in case 1. We could not ensure a specified DC gain in this range, the restriction made using the DC gain is more severe.

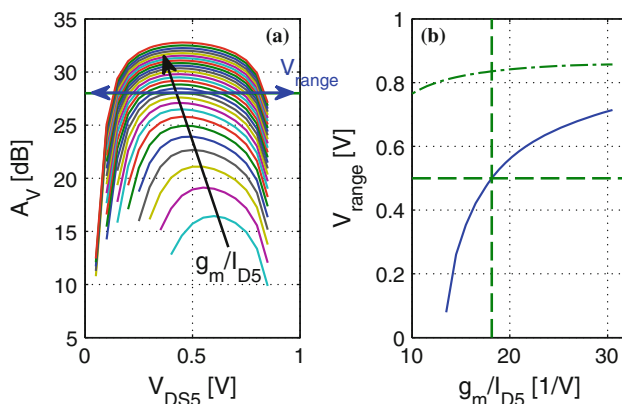


Fig. 14 (a) DC gain of stage Fig. 12(c) versus output DC voltage, for g_m/I_D of transistor $M5$ between 3 and 30 S/A, g_m/I_D of transistor $M6$ of 15 S/A and $L_5 = L_6 = 600$ nm; (b) voltage range where said DC gain is superior to $A_{V_0} > A_{V_{0,spec}}$ (solid blue) and output swing where $M5$ and $M6$ remain saturated (axis green) versus g_m/I_D of transistor $M5$ (Color figure online)

According to the specifications we target, one method or the other can be used.

4.2.3 Sizing from a small-signal specification (GBW, ϕ , Fig. 12(d))

Finally, if the specification to be insured is a small-signal characteristic, a $\{I_D, g_m/I_D, V_{DS}, L\}$ window can be found assuming a minimum gain-bandwidth product (GBW), transition frequency or phase margin (ϕ). In this last case, the sizing is no longer current independent, we need to add the current bias so as to find W but the transfer function can still be computed versus the bias current. With the notations of Fig. 12(d), the voltage gain in Laplace domain is

$$A_V(s) = \frac{C_{dg5} \cdot s - g_{m5}}{(C_L + C_{dd5}) \cdot s + g_{d5}} \quad (4)$$

The actual gain and phase shift and then DC open-loop gain, transition frequency and poles and zeros frequencies can be deduced versus DC power consumption. As an example, we can make a constant GBW sizing, finding g_m/I_D from GBW and the load capacitance, as a function of the drain current:

$$GBW = \frac{g_{m5}}{2\pi \cdot (C_L + C_{dd5})} = \frac{g_m/I_D|_5}{2\pi \cdot (C_L + C_{dd5})} \cdot I_D \quad (5)$$

Figure 15 shows the Bode diagram of the common-source stage Fig. 12(d) with a 600 nm-long standard- V_T NMOS, targeting a 1 MHz GBW on a 1 pF load capacitor, for several power consumption values.

The bias current is then set by a minimum DC gain or a phase shift value (see Fig. 16). For example, a minimum DC gain value of 30 dB requires at most 940 nA of bias current, that is a minimum g_m/I_D of 6.7 S/A.

In this section, we have seen that we can link circuit level specifications with the transistor biases and sizes through g_m/I_D , V_{DS} and L . We shall attempt to demonstrate

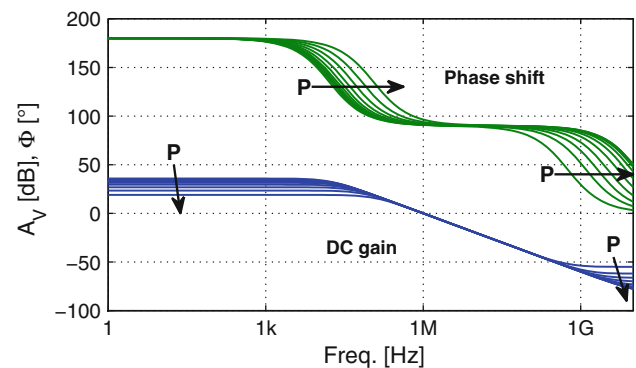


Fig. 15 Bode diagram, gain (blue) and phase (green), of stage Fig. 12(d), for several power consumption values between 250 nW and 2 μ W, constant g_{m5} , $V_{DD} = 1$ V, $C_L = 1$ pF (Color figure online)

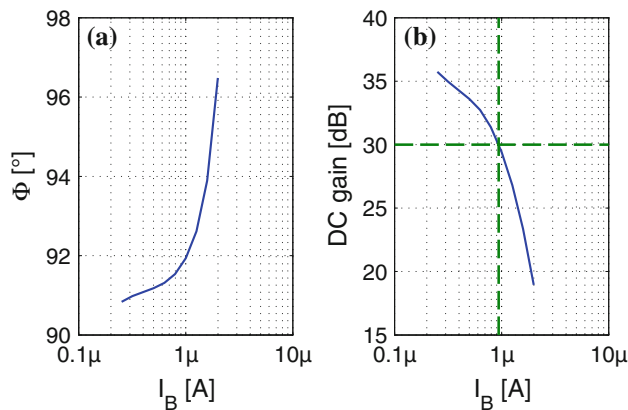


Fig. 16 Phase margin **a** and DC gain **b** versus consumed current, extracted from the Bode diagrams plotted in Fig. 15

in the next section the accuracy of this methodology on a double-stage amplifier.

5 Application to a single-ended Miller OTA

What has been done so far for simple cells can be extended to larger circuits. In that case however, the small-signal characteristics (transfer function— TF , common-mode or power supply rejection ratio— $CMRR$ or $PSRR$) become harder to calculate by hand. A symbolic calculator can then be used, implementing the matrix representation of a four-pole MOS transistor (see below) in order to compute a symbolic representation of the considered characteristic.

$$\begin{bmatrix} i_d \\ i_g \\ i_s \\ i_b \end{bmatrix} = \begin{bmatrix} C_{dd} & -C_{dg} & -C_{ds} & -C_{db} \\ -C_{gd} & C_{gg} & -C_{gs} & -C_{gb} \\ -C_{sd} & -C_{sg} & C_{ss} & -C_{sb} \\ -C_{bd} & -C_{bg} & -C_{bs} & C_{bb} \end{bmatrix} \cdot \begin{bmatrix} v_d \\ v_g \\ v_s \\ v_b \end{bmatrix} \cdot s + \begin{bmatrix} g_d & g_m & g_{ms} & g_{mb} \\ 0 & 0 & 0 & 0 \\ -g_d & -g_m & -g_{ms} & -g_{mb} \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} v_d \\ v_g \\ v_s \\ v_b \end{bmatrix} \quad (6)$$

We are considering a single-ended cascoded-bias Miller-compensated differential transconductance amplifier as depicted (Fig. 17). We shall imagine a particular sizing process for this circuit, and compare the sizing results with SPICE simulations.

5.1 Sizing preparation

The sizing process starts with the identification of all degrees of freedom and possible specifications according to the blocks of the chosen architecture. As degrees of freedom, we have the sizes of the differential pair ($W_1; L_1$), of

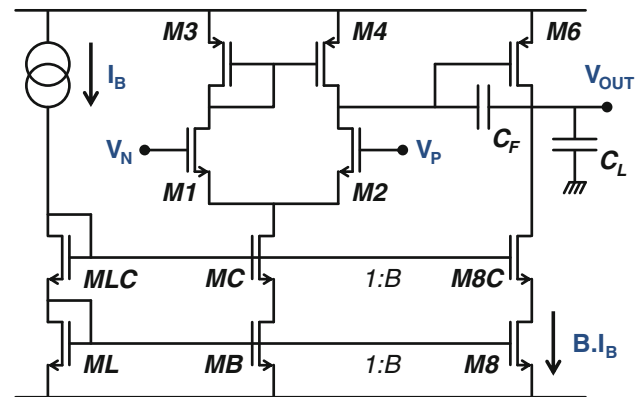


Fig. 17 Single-ended cascoded-bias Miller-compensated differential transconductance amplifier used as test vehicle

the load transistors ($W_3; L_3$), of the bias ($W_B; L_B$) and cascode ($W_C; L_C$) transistors, the current ratio between the first and second stage (B), the current bias (I_B) and the Miller capacitance value (C_F). Possible specifications include supply voltage (V_{DD}), maximum DC supply power (P_{MAX}), output swing ($V_{OUT,sw}$), input common-mode voltage range ($V_{CM,min} - V_{CM,max}$), DC gain, phase margin, $PSRR$, $CMRR$, systematic offset (V_{OFF}), load capacitance (C_L), input capacitance (C_{IN}), slew rate (SR), etc.

In DC domain, a possible systematic unbalance of this amplifier resides in the first stage: a difference in the DC voltage at $M3$ and $M4$ drains will lead to a degraded current copy in the current mirror. Having $V_{GS_6} = V_{GS_4}$ will then reduce the systematic offset, condition easily translated to having the same normalized current for the three transistors $M3, M4$ and $M6$, which explains the choice of $W_6 = 2B \cdot W_3$.

We then need to find a value for each degree of freedom by linking them through a set of $\{g_m/I_D, V_{DS}, L\}$ for each transistor to the specifications of this particular cell with a particular objective.

Starting with a priori guesses for B and the lengths (which will be refined later in the process), six degrees of freedom remain, we therefore need six specifications to deduce them from, along with the remaining possible specifications. Sizing processes in order to reach a certain DC gain or GBW were already introduced, we shall here target as an example:

- to optimize power consumption (1) for a given supply voltage ($V_{DD} = 1.0$ V),
- to encompass a certain range on the input common-mode voltage ($V_{CM,min} = 0.35$ V, $V_{CM,max} = 0.65$ V) (2 and 3) and maximize the output swing ($V_{OUT,sw}$) (4) around a nominal common-mode voltage ($V_{CM} = 0.5$ V) (5),
- to reach a minimum phase margin value ($\phi = 60^\circ$) (6), from which shall be inferred the DC gain, bandwidth on a 1 pF load capacitance, slew rate, $PSRR$ and $CMRR$.

To calculate these values, the OTA AC transfer function is needed. We may use two methods to evaluate it: considering only the most significant MOSFET parasitic capacitances (C_{gg} , C_{dg} , C_{gd} and C_{dd}), the basic conductances (g_m and g_d), and compute it by hand, or use the full matrix for each transistor (Eq. 6), and compute the TF using a symbolic solver.

Using the second method, to each AC node are associated a voltage and a current equation. All V_i of all transistors are replaced by their corresponding node voltage. Inputting the resulting current matrix in the node equations, all node voltages shall be found, among them V_{OUT} , as a function of V_N , V_P and V_{DD} . We then get V_{OUT}/V_N , V_{OUT}/V_P and V_{OUT}/V_{DD} by nullifying the other voltages, and find the TF , $CMRR$, and $PSRR$ as a function of all capacitances and conductances.

Symbolic calculation is more precise and faster to compute, its precision only depending on the respective accuracy of both calculators, SPICE and Matlab, whereas the other method precision depends on the legitimacy of the approximations on transconductances and capacitances. For short transistors, g_d can be significant compared to g_m , and parasitic capacitances all have the same order of magnitude.

5.2 Sizing process

As we said, for each transistor, g_m/I_D and V_{DS} are searched for, unlike the other g_m/I_D methodologies where V_{DS} effects were neglected.

Figure 18 presents the sizing sequence used in the process. We start by maximizing the output swing ($V_{OUT,sw} = V_{DD} - (V_{DS8} + V_{Dsat8C} + V_{Dsat6})$). As $V_{GS8} = V_{DS8}$ to get a good current copy, we can start minimizing the gate-source voltage of $M8$. From this, and knowing the output DC voltage, V_{DS8C} is known, and g_m/I_D is found by minimizing V_{Dsat8C} . M_B is a copy of $M8$, considering equal their V_{GS} and V_{DS} , but with different bias currents. M_C has the same gate-source voltage as $M8C$, with $V_{DS_C} = V_{CM} - V_{DS8} - V_{GS1} = V_{CM} - V_{CM,min} + V_{Dsat8C}$. The minimum input common-mode voltage then determines $V_{GS1} = V_{CM,min} - V_{DS_B} - V_{Dsat8C}$. $M1$ drain-source voltage is given through the maximum input common-mode voltage $V_{DS1} = V_{CM,max} + V_{Dsat1} - V_{DS_B} = V_{DS_C} - V_{GS1}$. $M2$ is a strict copy of $M1$. Using the maximum common-mode voltage required, and $M2$ characteristics, $M6$ is sized from its $V_{GS} = V_{DD} - V_{CM,max} + V_{GS2} - V_{Dsat2}$. $M3$ and $M4$ are finally sized considering their drain-source and gate-source voltages equal to V_{GS6} .

At this time, all transistors g_m/I_D , V_{DS} and L are set, from which are derived all characteristics as a function of the power consumption. We may then compute the complete

transfer function for different values of the Miller capacitor, and select for a given power consumption the capacitor value that bestows a phase margin of 60° (cf. Fig. 19).

Having the capacitor value, the transition frequency and therefore the design power efficiency (GBW/P) may be extracted versus the power consumption. Those two values are plotted in Fig. 20 considering for the efficiency only designs the widths of which are in the technology's [W_{MIN} , W_{MAX}] range.

Choosing the power consumption so that the power efficiency is maximum, all other specifications can be

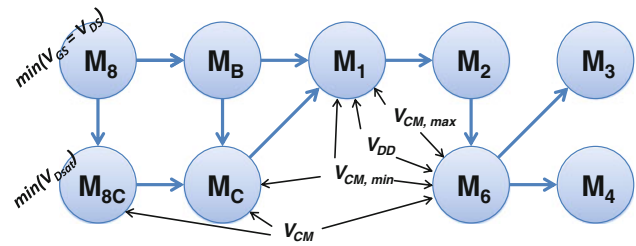


Fig. 18 Transistors sizing sequence for the amplifier of Fig. 17

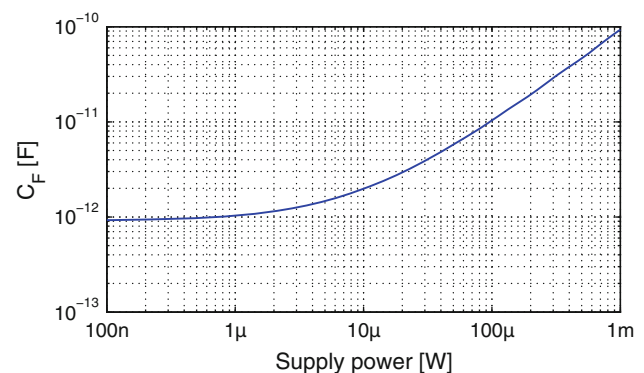


Fig. 19 Miller capacitance needed for a 60° phase margin versus power consumption

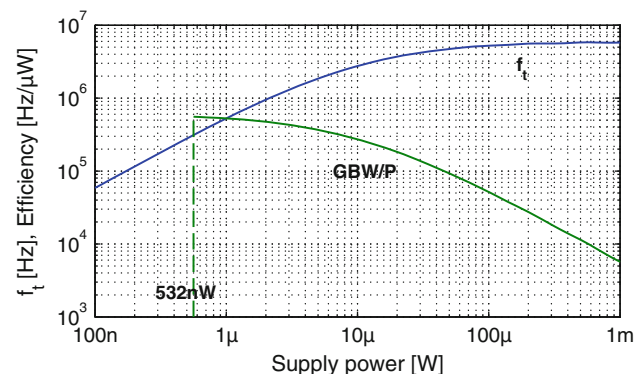


Fig. 20 Transition frequency (solid blue) and power efficiency (GBW/P) (solid green) of the amplifier versus power consumption (Color figure online)

computed. According to the results, we may loop using different values of lengths or of the current multiplier B (example Fig. 21), originally guessed arbitrarily.

5.3 Results

So as to ascertain the accuracy of our sizing methodology, a comparison between SPICE and Matlab outputs shall be carried out. All transistors sizes and the bias current (I_B) found using the Matlab script are input to the SPICE netlist describing the test schematic (Fig. 22). The device under test (DUT) is stimulated with a unity-input AC signal, around the calculated common-mode voltage. If both DC input voltages were forced though, the amplifier output would saturate trying to reach $V_{OFF} \cdot A_{V_0}$. The negative feedback loop with infinite inductor L_{DCFEED} and capacitor C_{DCCUT} is made so the DC differential input stabilizes at the offset voltage, while the amplifier remains in open-loop for AC signals.

Table 2 sums up the sizes, parameters and errors on each transistor parameter between the Matlab estimation and the circuit simulated using this test schematic. Keeping in mind the remarks on g_{mb} made earlier, the listed values remain highly accurate.

Figure 23 plots the error on the transfer function Bode diagram, the gain error does not go beyond 0.3 dB, the phase error, beyond 2.3° . Table 3 lists the errors on the target specifications at circuit level. The $CMRR$ is simulated using the test schematic with in-phase input signals.

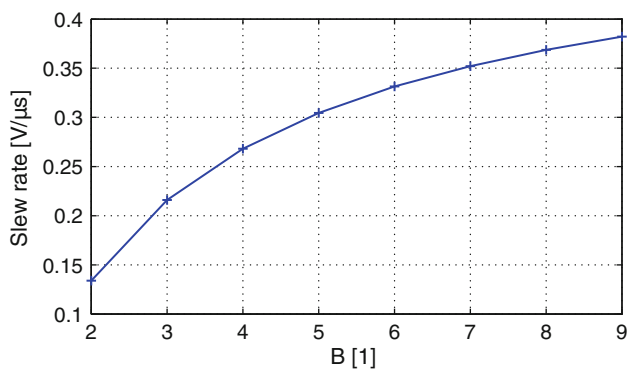


Fig. 21 Slew rate of the OTA versus current multiplier B , constant power consumption

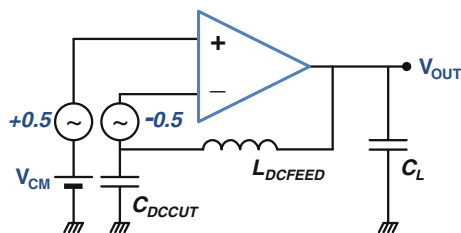


Fig. 22 SPICE AC analysis schematic (units are then arbitrary)

Table 2 Error on transistor characteristics between Matlab estimation and SPICE simulation

Values	g_m/I_D	V_{DS}	W	L	
	(S/A)	(V)	(μm)	(μm)	
<i>M1</i>	29.2	0.19	1.60	0.60	
<i>M2</i>	29.2	0.19	1.60	0.60	
<i>M3</i>	26.2	0.46	2.30	0.42	
<i>M4</i>	26.2	0.46	2.30	0.42	
<i>M6</i>	26.2	0.50	9.20	0.42	
<i>M8</i>	29.1	0.15	6.70	0.60	
<i>M8C</i>	21.3	0.35	0.24	0.30	
<i>MB</i>	29.1	0.15	3.30	0.60	
<i>MC</i>	21.6	0.20	0.12	0.30	
Errors	g_m/I_D	V_{GS}	V_{DS}	V_{Dsat}	
	(S/A)	(mV)	(mV)	(mV)	
<i>M1</i>	0.01	2.3	12.9	0.1	
<i>M2</i>	0.01	2.7	20.6	0.1	
<i>M3</i>	0.03	16.4	16.4	0.2	
<i>M4</i>	0.03	16.4	24.0	0.2	
<i>M6</i>	0.05	24.0	0.4	0.2	
<i>M8</i>	0.15	3.8	16.1	0.2	
<i>M8C</i>	0.04	23.6	15.7	3.5	
<i>MB</i>	0.07	3.7	1.8	0.1	
<i>MC</i>	0.08	41.6	5.3	5.3	
Errors	g_m	g_d	g_{ms}	g_{mb}	C_m
	(%)	(%)	(%)	(%)	(%)
<i>M1</i>	0.3	3.1	0.1	1.0	1.2
<i>M2</i>	0.2	5.9	0.1	1.1	2.2
<i>M3</i>	0.5	7.1	0.7	6.3	2.2
<i>M4</i>	0.3	8.4	0.8	6.4	2.5
<i>M6</i>	5.4	8.9	3.6	5.1	2.2
<i>M8</i>	5.0	4.2	4.3	1.7	2.8
<i>M8C</i>	5.3	8.6	6.3	15.4	3.7
<i>MB</i>	0.0	0.4	0.2	2.8	1.0
<i>MC</i>	0.6	0.6	2.4	23.2	6.7

The limited error made on the transistor parameters impacts to a small enough extent the circuit level parameters. We demonstrated in this section that we can calculate all the parameters of a circuit from specifications and link them, through g_m/I_D , V_{DS} and L , to the transistor sizes and bias voltages.

6 Corners dependence and technology transfer

An estimation of all the circuit parameters in process (SS, TT, FF), voltage (0.9, 1.0, 1.1 V) and temperature (-25 ,

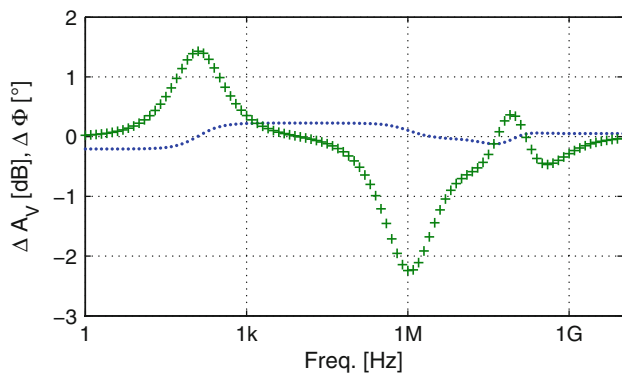


Fig. 23 Error on Bode diagram, gain (blue dots) and phase (green crosses), between Matlab estimation and SPICE simulation (Color figure online)

Table 3 Error on circuit level characteristics between Matlab estimation and SPICE simulation

	Values	Errors
P_{DD}	532 nW	2.8 %
I_B	133 nA	0.3 %
I_8	266 nA	5.6 %
GBW	296 kHz	0.2 %
f_t	296 kHz	0.1 %
$Slew\ rate$	134 V/ms	1.1 %
ϕ	60.0°	1.6°
A_{V_0}	67.1 dB	0.2 dB
$CMRR_0$	66.1 dB	0.6 dB
$V_{CM,min}$	350 mV	9.8 mV
$V_{CM,max}$	650 mV	21.4 mV
$V_{OUT,sw}$	756 mV	12.4 mV

25, 75 °C) corners (PVT) can be carried out, by extracting the transistors characteristics in a new database for each considered corner.

The results plotted in Figs. 24 and 25 were obtained with Matlab considering I_B constant, with the sizes listed in Table 2. We can hardly assume a priori which corner will be the worst, for example, worst DC gain is for FF, 0.9 V, 75 °C, worst V_{CM} for FF, 0.9 V, −25 °C, worst stability for TT, 1.1 V, −25 °C. The choice of the current reference (constant I_B , constant g_{m1} or constant g_m/I_{D1}) will highly impact the results as well. The choice of the corners to consider is in the designer's hand.

In the same way, this methodology allows quick adaptation from a technology to another: with exactly the same specifications, the OTA has been transferred to a 130 nm CMOS technology, only changing the input technology files of the script. The sizing recalculations are performed with Matlab in a few seconds of CPU time. Table 4 sums up the errors made. The specifications are the same as the

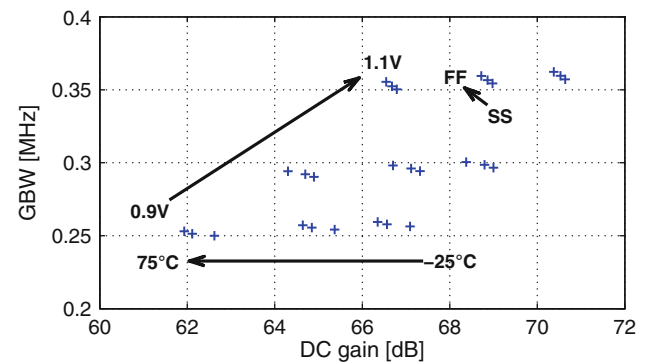


Fig. 24 DC gain versus gain-bandwidth product variation with PVT corners (process SS, TT, FF, voltage 0.9, 1.0, 1.1 V and temperature −25, 25, 75 °C)

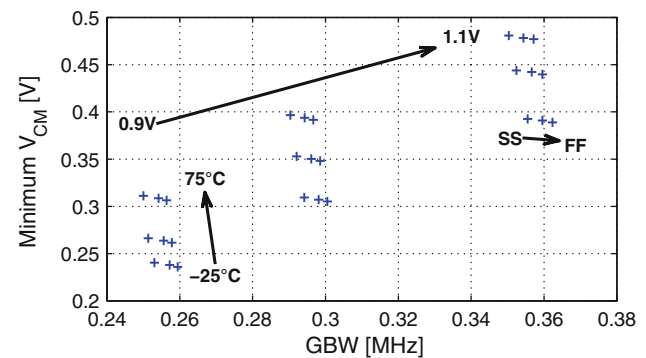


Fig. 25 Minimum common-mode voltage versus gain-bandwidth product variation with PVT corners (process SS, TT, FF, voltage 0.9, 1.0, 1.1 V and temperature −25, 25, 75 °C)

Table 4 Error on circuit level characteristics between Matlab estimation and SPICE simulation for a 130 nm bulk technology

	Values	Errors
P_{DD}	532 nW	0.1 %
I_B	133 nA	0.1 %
I_8	266 nA	0.3 %
GBW	35 kHz	0.5 %
f_t	40 kHz	1.7 %
$Slew\ rate$	35 V/ms	7.0 %
ϕ	60.0°	1.5°
A_{V_0}	80.1 dB	0.5 dB
$CMRR_0$	83.4 dB	1.0 dB
$V_{CM,min}$	350 mV	0.2 mV
$V_{CM,max}$	650 mV	2.6 mV
$V_{OUT,sw}$	675 mV	2.1 mV

last design, though of course the other performances change according to the technology. The sizing technique remains just as effective.

7 Conclusion

The original g_m/I_D methodology (Sect. 2) was based on the assumption that the g_m/I_D versus I_N characteristic was independent of size and drain biasing. Approximations were made to calculate the other transistor characteristics (proportional-to-length V_{EA} , equations for capacitors, DC and large signal values). In Sect. 3 we demonstrated that these assumptions and relations were accurate enough for long-channel transistors but could hardly be used for short-channel transistors of advanced CMOS technologies. The database upon which a new sizing technique can be developed has to take into account the effects of length and drain-source voltage on the MOS characteristics in addition to the inversion level.

Section 4 introduced the updated sizing technique, ensuring its effectiveness for a single transistor, then explaining the way to link circuit level specifications with transistors g_m/I_D , V_{DS} and L .

The application Sect. 5 uses a double-stage amplifier as test vehicle and details the sizing and optimization process before comparing the results accuracy with SPICE simulation. Using alternative databases, in Sect. 6, the methodology can estimate corner variations of all circuit parameters. Using a database generated out of a 130 nm technology, the same amplifier was designed automatically with the same specifications on another technology with no new design effort in a very short time.

We have demonstrated the effectiveness of our sizing methodology. It is quick, accurate, technology independent, and should be applicable to any more advanced CMOS technology.

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